

4.10.4 The Haber process and the use of NPK fertilisers (chemistry only)

4.10.4.1 The Haber process

Content	Key opportunities for skills development
The Haber process is used to manufacture ammonia, which can be used to produce nitrogen-based fertilisers. The raw materials for the Haber process are nitrogen and hydrogen. Students should be able to recall a source for the nitrogen and a source for the hydrogen used in the Haber process. The purified gases are passed over a catalyst of iron at a high temperature (about 450°C) and a high pressure (about 200 atmospheres). Some of the hydrogen and nitrogen reacts to form ammonia. The reaction is reversible so some of the ammonia produced breaks down into nitrogen and hydrogen: <i>nitrogen + hydrogen $\rightleftharpoons$ ammonia</i> On cooling, the ammonia liquefies and is removed. The remaining hydrogen and nitrogen are recycled.	MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios, fractions and percentages.
(HT only) Students should be able to: interpret graphs of reaction conditions versus rate	MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios, fractions and percentages.
- apply the principles of dynamic equilibrium in Reversible reactions and dynamic equilibrium (page 59) to the Haber process - explain the trade-off between rate of production and position of equilibrium - explain how the commercially used conditions for the Haber process are related to the availability and cost of raw materials and energy supplies, control of equilibrium position and rate.	WS 3.5, 3.8

4.10.4.2 Production and uses of NPK fertilisers

Content	Key opportunities for skills development
Compounds of nitrogen, phosphorus and potassium are used as fertilisers to improve agricultural productivity. NPK fertilisers contain compounds of all three elements. Industrial production of NPK fertilisers can be achieved using a variety of raw materials in several integrated processes. NPK fertilisers are formulations of various salts containing appropriate percentages of the elements. Ammonia can be used to manufacture ammonium salts and nitric acid. Potassium chloride, potassium sulfate and phosphate rock are obtained by mining, but phosphate rock cannot be used directly as a fertiliser. Phosphate rock is treated with nitric acid or sulfuric acid to produce soluble salts that can be used as fertilisers. Students should be able to: • recall the names of the salts produced when phosphate rock is treated with nitric acid, sulfuric acid and phosphoric acid • compare the industrial production of fertilisers with laboratory preparations of the same compounds, given appropriate information.	AT 4 Prepare an ammonium salt.